AIC ACT policy training

Weights & Biases training report · finished runs only

Projects Imperial-College-London-SPQR/aic-act-training · Imperial-College-London-SPQR/aic-act-policy

1 Overview: finished runs

Project	Run	W&B id	Created (UTC)	dataset.repo_id	Steps logged / set	train/epochs	train/loss	train/l1_loss	train/kld_loss	min train/loss	Runtime
aic-act-training	experiment_8_act_2026-04-13_23-28-27	j6vqaecl	2026-04-13 22:28	experiment_8_three_camera_images	200 / 20000	6.3	0.0249	0.01291	1.45e-03	0.0249	11 min
	sfp_clean_198_2026-04-20_14-06-27	s4d32f30	2026-04-20 13:06	sfp_clean_198_2026-04-20_12-31-19	200 / 20000	0.9	0.05815	0.02553	1.67e-03	0.05483	9 min
	sfp_clean_198_2026-04-20_14-20-08	uypylo0g	2026-04-20 13:20	sfp_clean_198_2026-04-20_12-31-19	2200 / 220000	10.0	0.01032	0.005573	3.90e-05	0.009257	1h 35m
	sfp_clean_246_2026-04-20_18-24-56	onesdtgp	2026-04-20 17:24	sfp_clean_198_plus_48_2026-04-20_16-37-48	2200 / 220000	8.1	0.01055	0.006437	4.67e-05	0.009481	1h 45m
	sfp_student_clean_497_48ep_2026-04-21_15-28-25	c7dpskzh	2026-04-21 14:28	sfp_student_clean_497_seed_6531465314	26400 / 2640000	95.7	0.003046	0.002654	1.21e-06	0.002663	23h 23m
	sc_stage_fixed_scene_curated_20260508_002_terminal_v1_guarded_insert_no_imagenet_2026-05-10_13-05-24	c0e9wq0x	2026-05-10 12:05	guarded_insert	200 / 20000	21.3	0.01681	0.01862	1.32e-04	0.01681	13 min
aic-act-policy	experiment_8_act_2026-04-12_15-55-44	q2xjpe2q	2026-04-12 14:55	experiment_8_center_camera	200 / 20000	6.3	0.02587	0.01479	1.45e-03	0.02548	8 min
	experiment_8_act_2026-04-13_22-10-30	j9hgfa4t	2026-04-13 21:10	experiment_8_three_camera_images	200 / 20000	6.3	0.02597	0.01297	1.37e-03	0.02502	10 min

2 Training curves: all finished runs (log step axis)

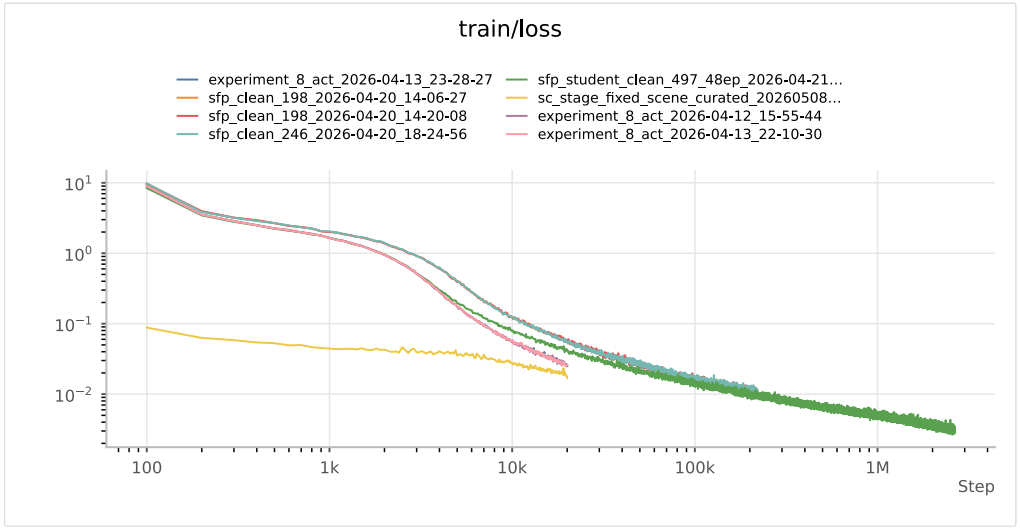


Fig. 1 train/loss

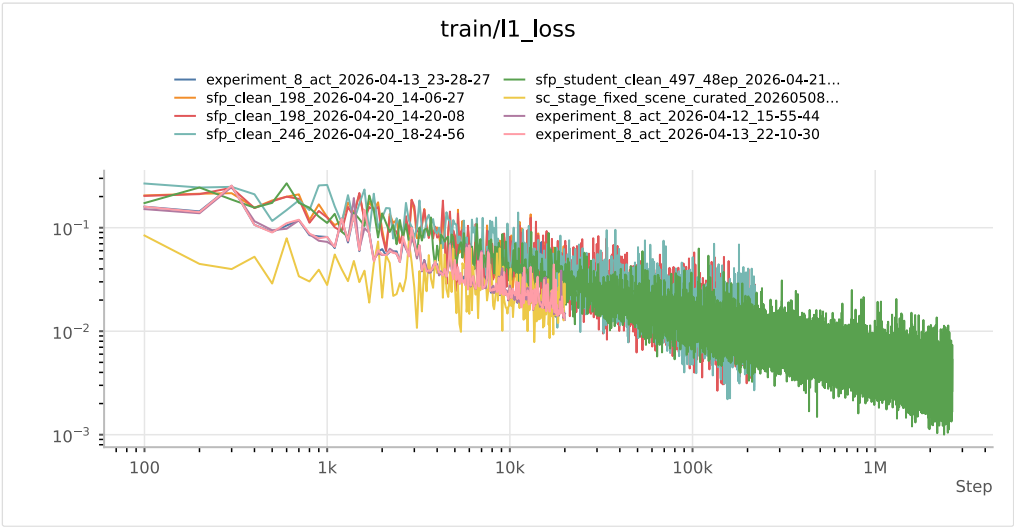


Fig. 2 train/l1_loss

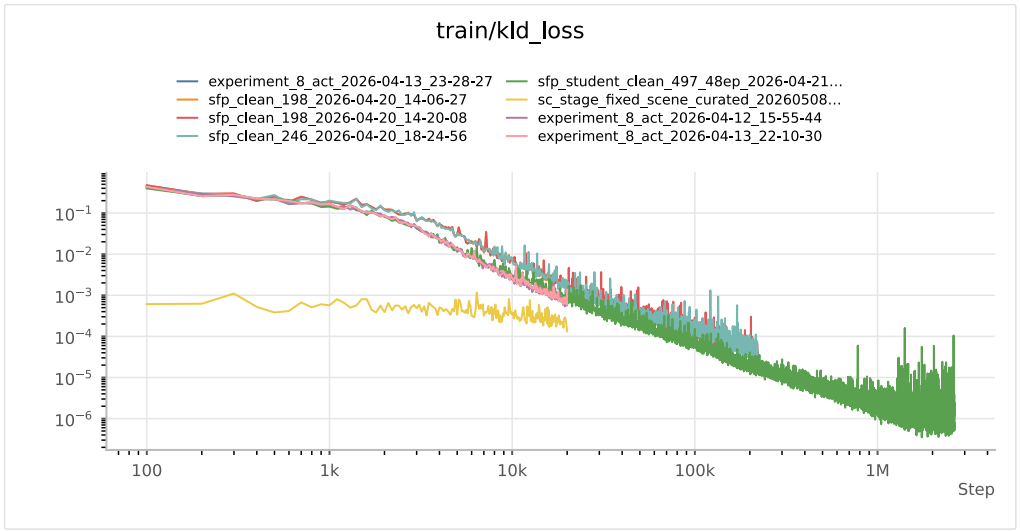


Fig. 3 train/kld_loss

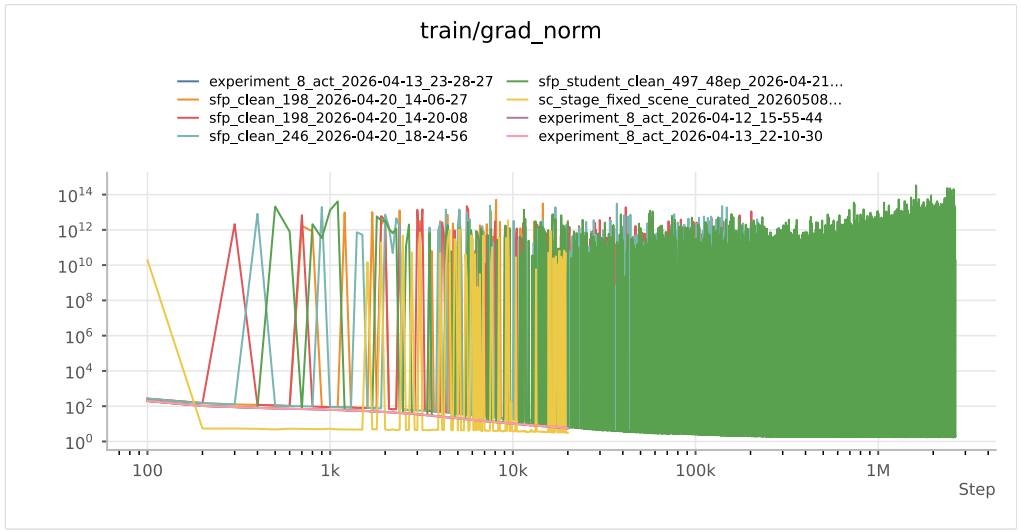


Fig. 4 train/grad_norm

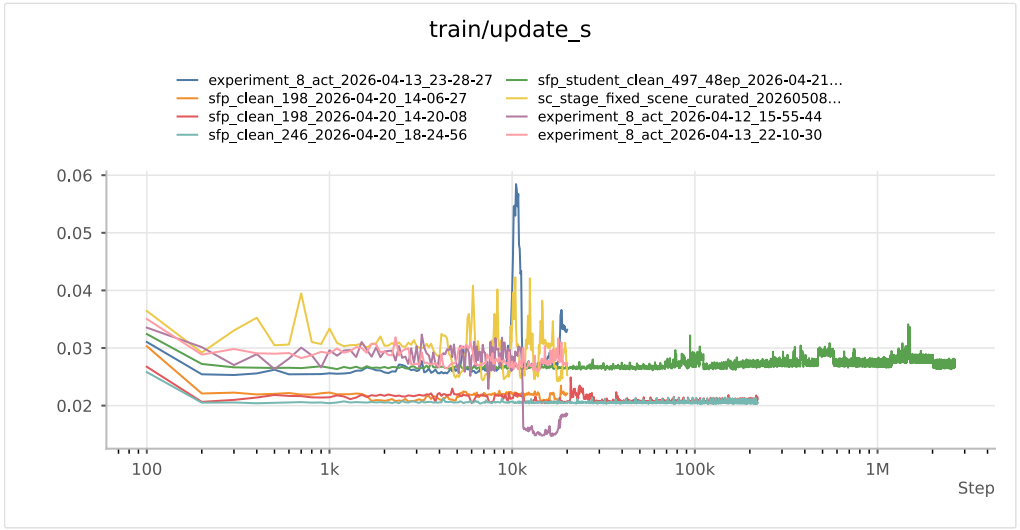


Fig. 5 train/update_s

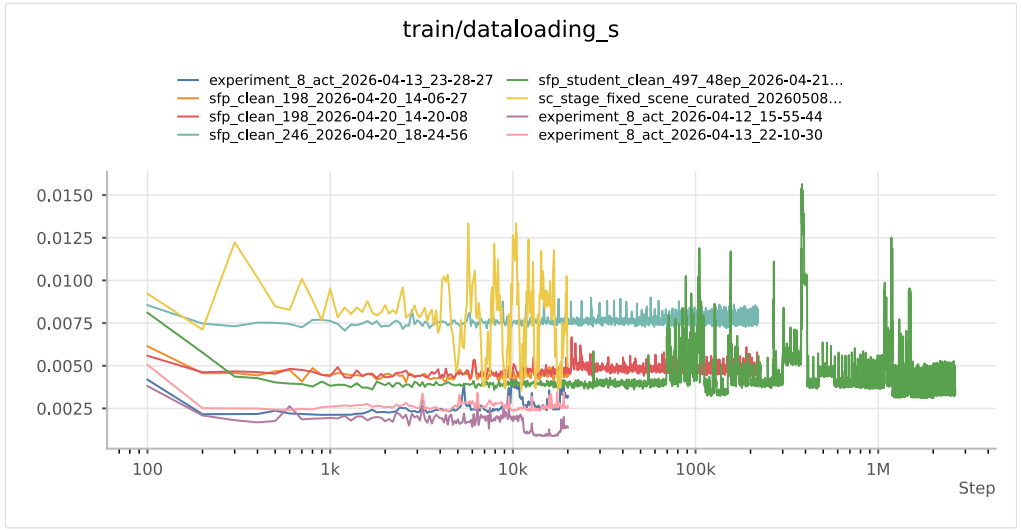


Fig. 6 train/dataloading_s

3 Training curves per run

experiment_8_act_2026-04-13_23-28-27

aic-act-training · 200 steps logged · final train/loss 0.0249

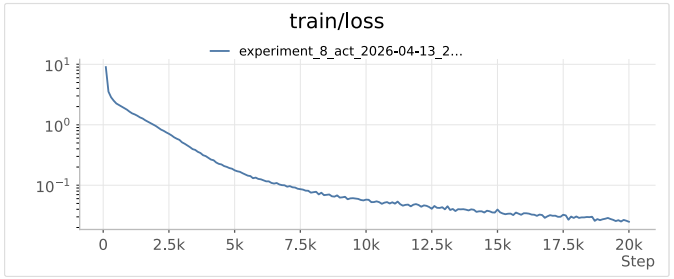


Fig. 7 experiment_8_act_2026-04-13_23-28-27 · train/loss

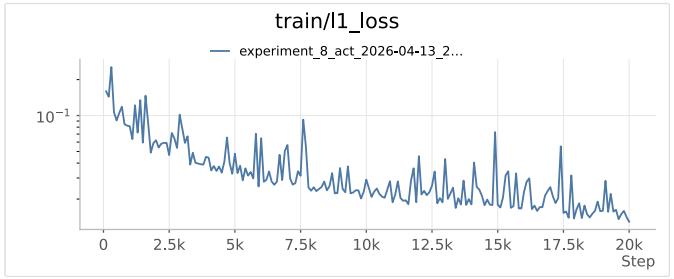


Fig. 8 experiment_8_act_2026-04-13_23-28-27 · train/l1_loss

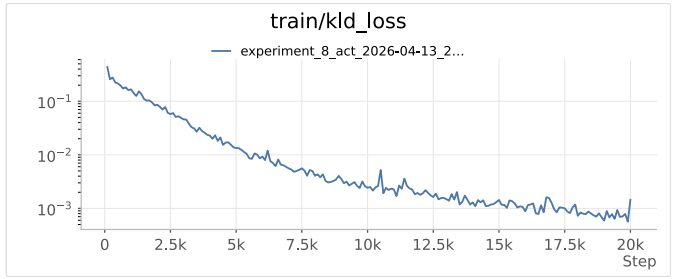


Fig. 9 experiment_8_act_2026-04-13_23-28-27 · train/kld_loss

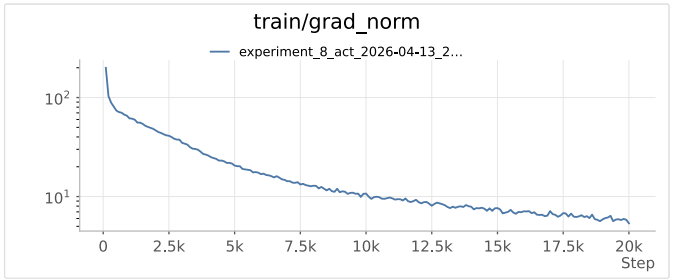


Fig. 10 experiment_8_act_2026-04-13_23-28-27 · train/grad_norm

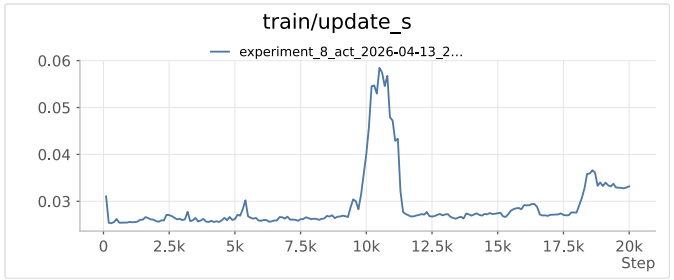


Fig. 11 experiment_8_act_2026-04-13_23-28-27 · train/update_s

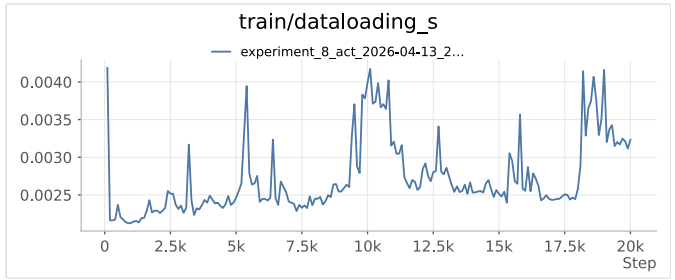


Fig. 12 experiment_8_act_2026-04-13_23-28-27 · train/dataloading_s

sfp_clean_198_2026-04-20_14-06-27

aic-act-training · 200 steps logged · final train/loss 0.05815

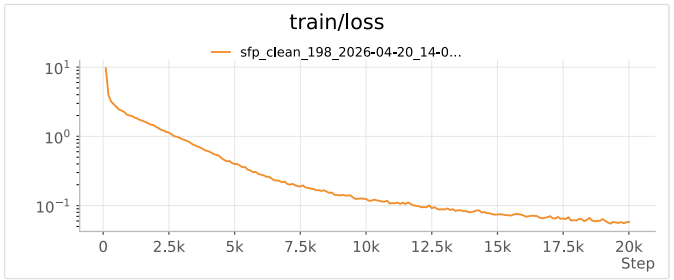


Fig. 13 sfp_clean_198_2026-04-20_14-06-27 · train/loss

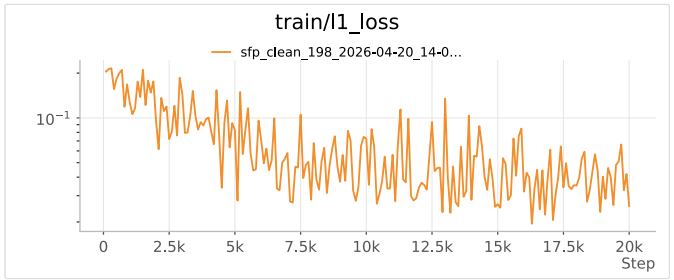


Fig. 14 sfp_clean_198_2026-04-20_14-06-27 · train/l1_loss

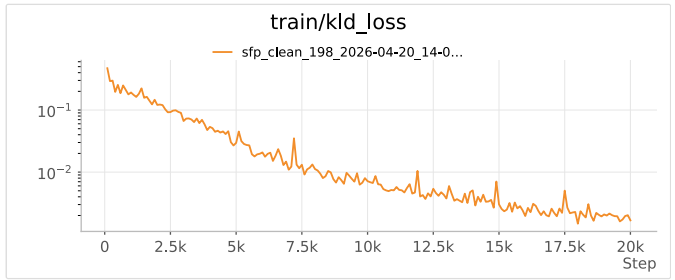


Fig. 15 sfp_clean_198_2026-04-20_14-06-27 · train/kld_loss

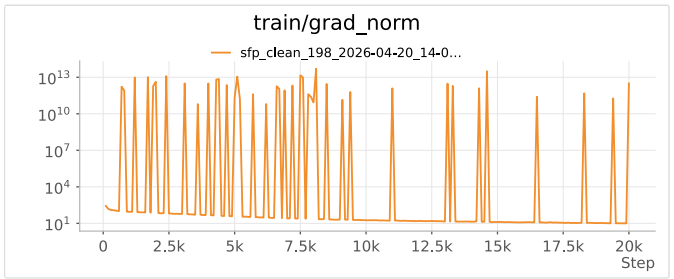


Fig. 16 sfp_clean_198_2026-04-20_14-06-27 · train/grad_norm

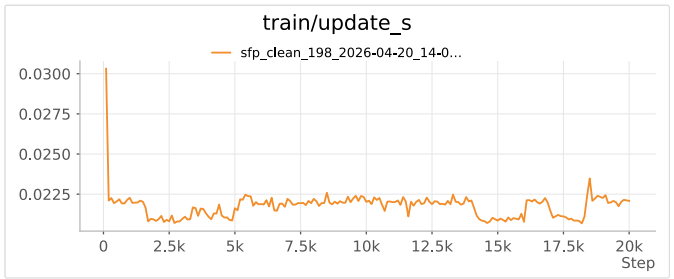


Fig. 17 sfp_clean_198_2026-04-20_14-06-27 · train/update_s

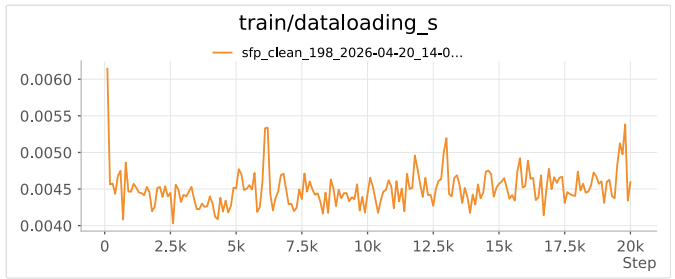


Fig. 18 sfp_clean_198_2026-04-20_14-06-27 · train/dataloading_s

sfp_clean_198_2026-04-20_14-20-08 aic-act-training · 2200 steps logged · final train/loss 0.01032

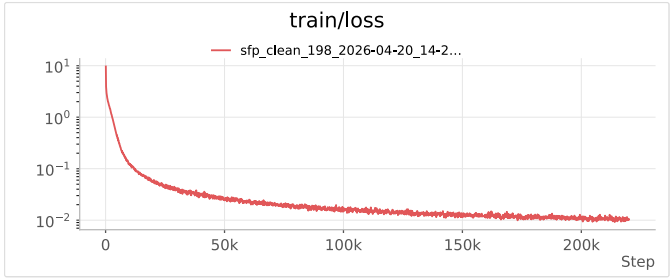


Fig. 19 sfp_clean_198_2026-04-20_14-20-08 · train/loss

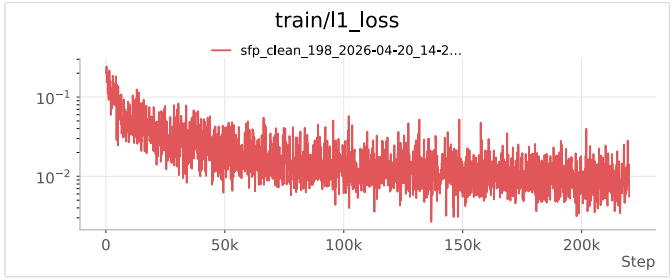


Fig. 20 sfp_clean_198_2026-04-20_14-20-08 · train/l1_loss

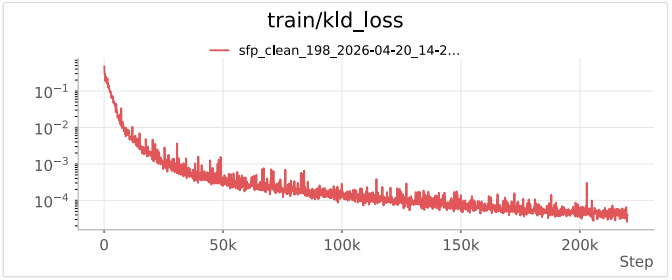


Fig. 21 sfp_clean_198_2026-04-20_14-20-08 · train/kld_loss

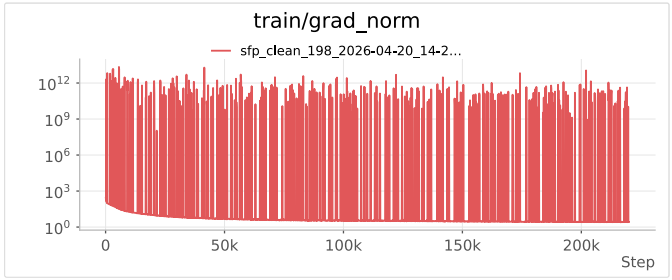


Fig. 22 sfp_clean_198_2026-04-20_14-20-08 · train/grad_norm

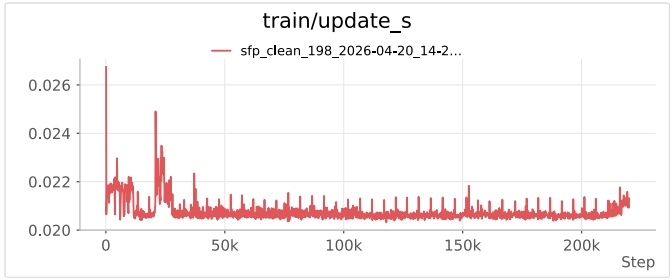


Fig. 23 sfp_clean_198_2026-04-20_14-20-08 · train/update_s

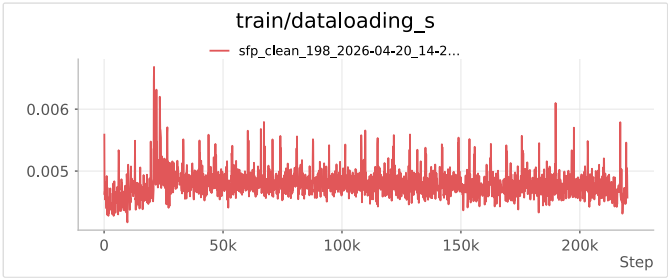


Fig. 24 sfp_clean_198_2026-04-20_14-20-08 · train/dataloading_s

sfp_clean_246_2026-04-20_18-24-56 aic-act-training · 2200 steps logged · final train/loss 0.01055

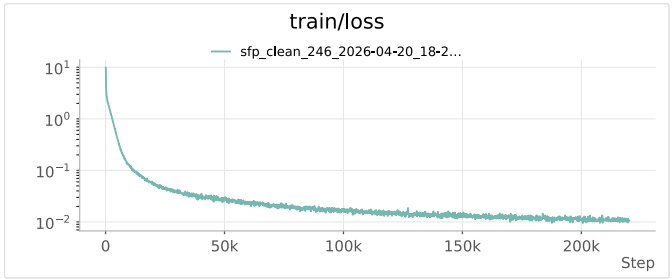


Fig. 25 sfp_clean_246_2026-04-20_18-24-56 · train/loss

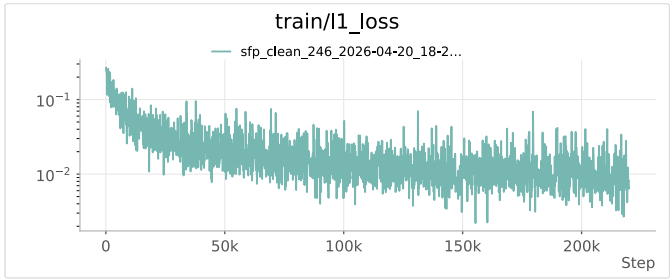


Fig. 26 sfp_clean_246_2026-04-20_18-24-56 · train/l1_loss

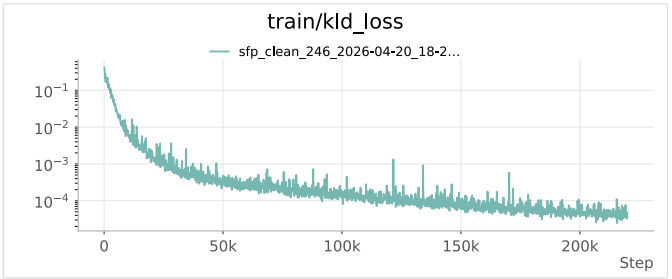


Fig. 27 sfp_clean_246_2026-04-20_18-24-56 · train/kld_loss

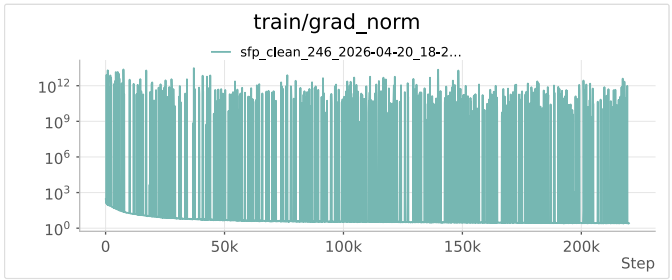


Fig. 28 sfp_clean_246_2026-04-20_18-24-56 · train/grad_norm

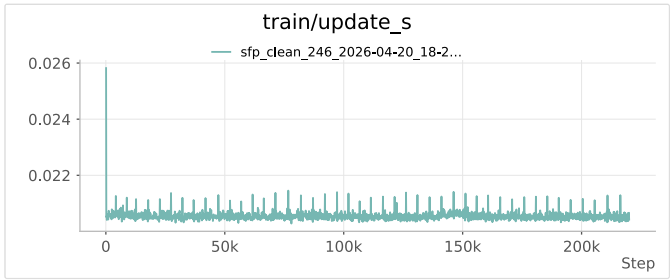


Fig. 29 sfp_clean_246_2026-04-20_18-24-56 · train/update_s

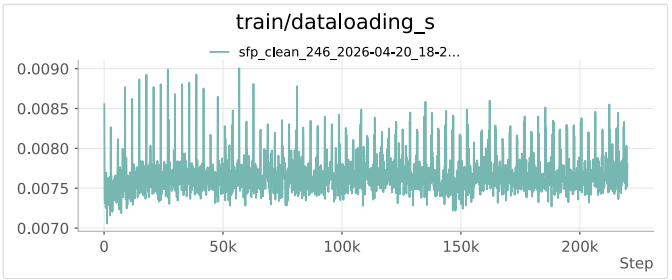


Fig. 30 sfp_clean_246_2026-04-20_18-24-56 · train/dataloading_s

sfp_student_clean_497_48ep_2026-04-21_15-28-25 aic-act-training · 26400 steps logged · final train/loss 0.003046

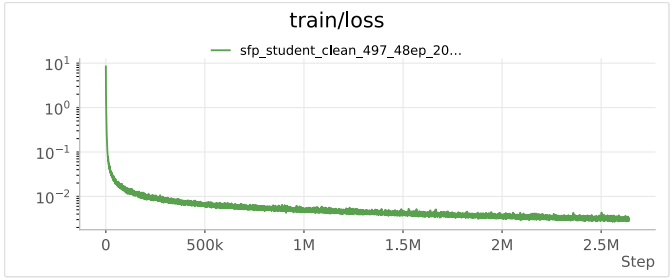


Fig. 31 sfp_student_clean_497_48ep_2026-04-21_15-28-25 · train/loss

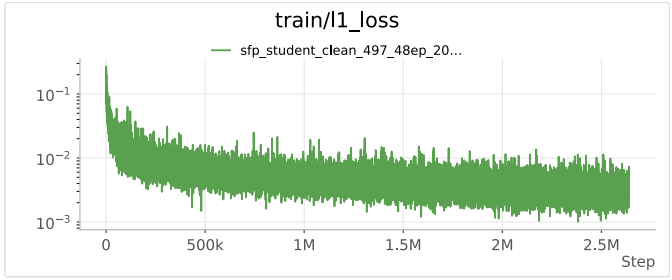


Fig. 32 sfp_student_clean_497_48ep_2026-04-21_15-28-25 · train/l1_loss

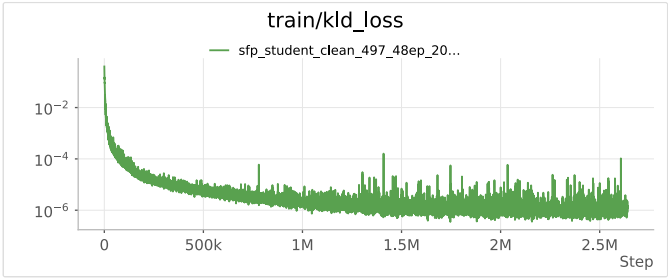


Fig. 33 sfp_student_clean_497_48ep_2026-04-21_15-28-25 · train/kld_loss

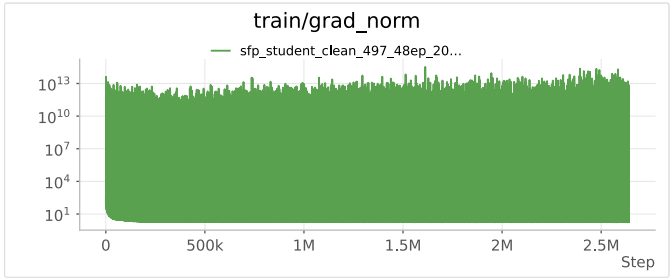


Fig. 34 sfp_student_clean_497_48ep_2026-04-21_15-28-25 · train/grad_norm

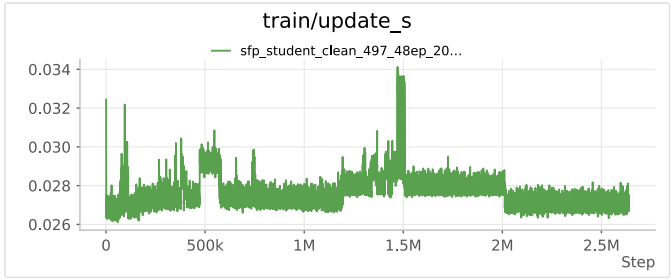


Fig. 35 sfp_student_clean_497_48ep_2026-04-21_15-28-25 · train/update_s

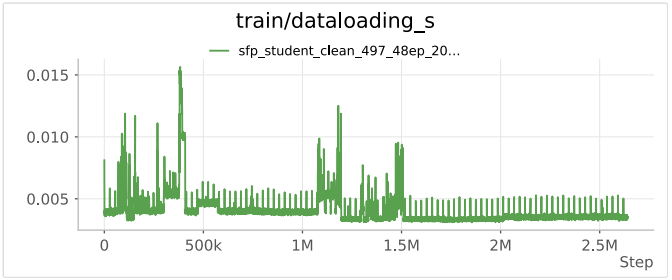


Fig. 36 sfp_student_clean_497_48ep_2026-04-21_15-28-25 · train/dataloading_s

sc_stage_fixed_scene_curated_20260508_002_terminal_v1_guarded_insert_no_imagenet_2026-05-10_13-05-24 aic-act-training · 200 steps logged · final train/loss 0.01681

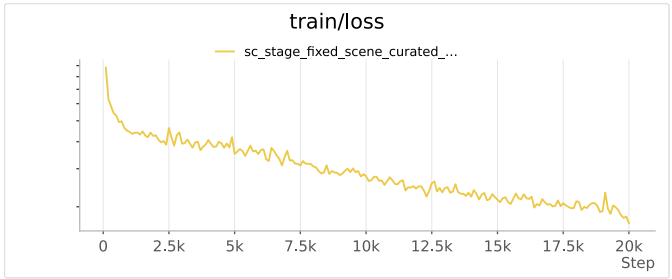


Fig. 37 sc_stage_fixed_scene_curated_20260508_002_terminal_v1_guarded_insert_no_imagenet_2026-05-10_13-05-24 · train/loss

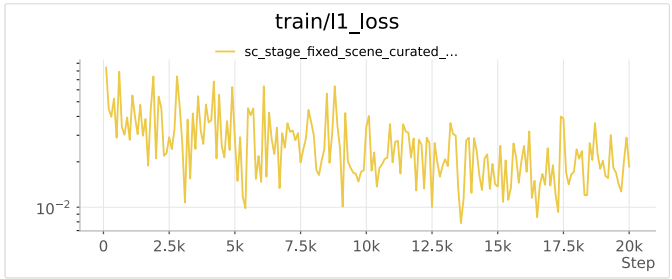


Fig. 38 sc_stage_fixed_scene_curated_20260508_002_terminal_v1_guarded_insert_no_imagenet_2026-05-10_13-05-24 · train/l1_loss

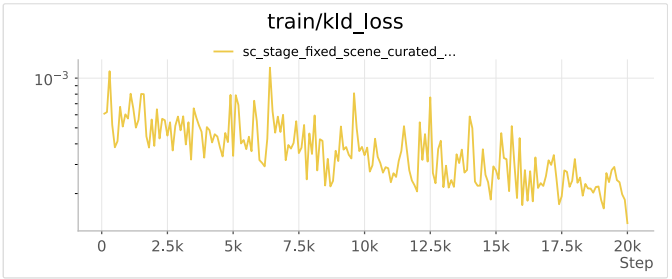


Fig. 39 sc_stage_fixed_scene_curated_20260508_002_terminal_v1_guarded_insert_no_imagenet_2026-05-10_13-05-24 · train/kld_loss

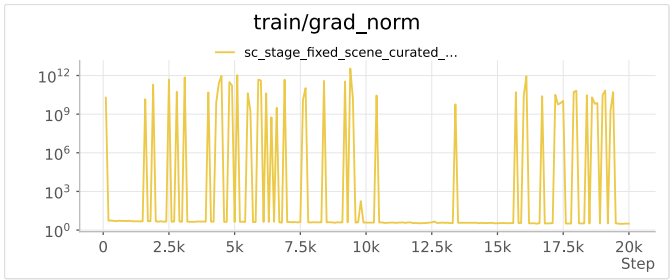


Fig. 40 sc_stage_fixed_scene_curated_20260508_002_terminal_v1_guarded_insert_no_imagenet_2026-05-10_13-05-24 · train/grad_norm

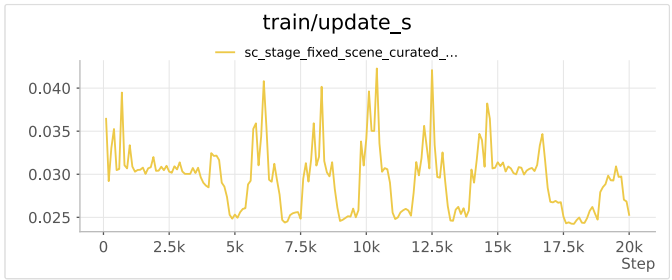


Fig. 41 sc_stage_fixed_scene_curated_20260508_002_terminal_v1_guarded_insert_no_imagenet_2026-05-10_13-05-24 · train/update_s

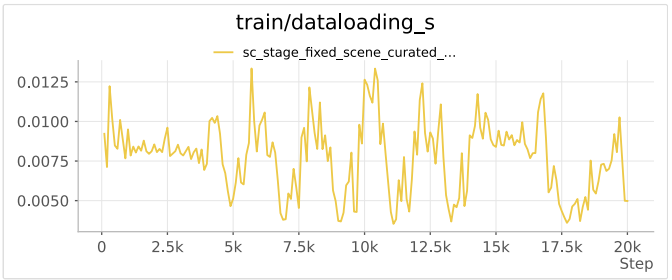


Fig. 42 sc_stage_fixed_scene_curated_20260508_002_terminal_v1_guarded_insert_no_imagenet_2026-05-10_13-05-24 · train/dataloading_s

experiment_8_act_2026-04-12_15-55-44 aic-act-policy · 200 steps logged · final train/loss 0.02587

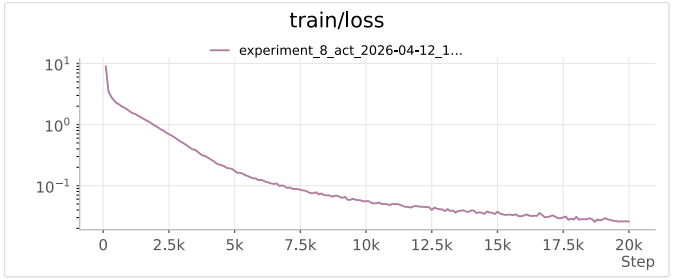


Fig. 43 experiment_8_act_2026-04-12_15-55-44 · train/loss

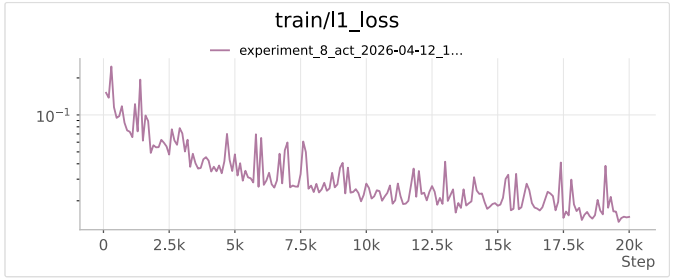


Fig. 44 experiment_8_act_2026-04-12_15-55-44 · train/l1_loss

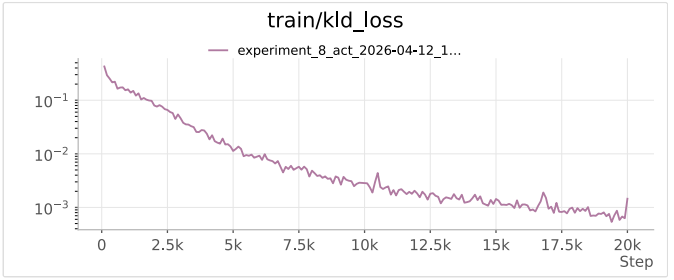


Fig. 45 experiment_8_act_2026-04-12_15-55-44 · train/kld_loss

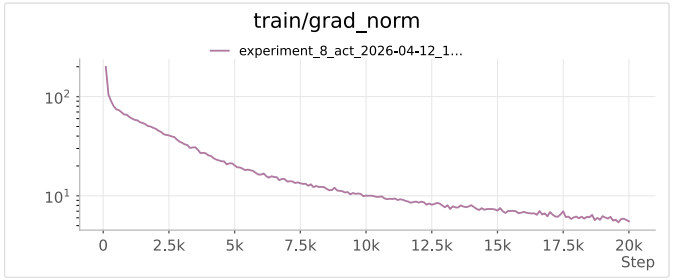


Fig. 46 experiment_8_act_2026-04-12_15-55-44 · train/grad_norm

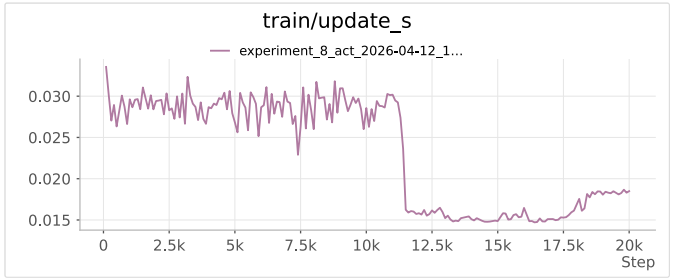


Fig. 47 experiment_8_act_2026-04-12_15-55-44 · train/update_s

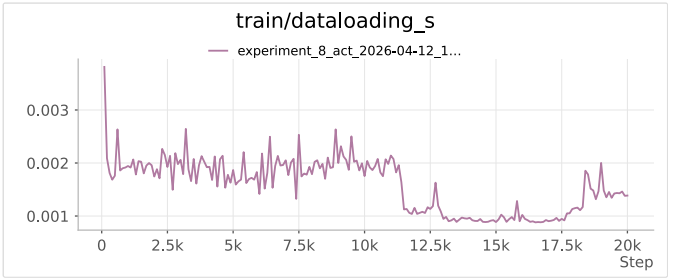


Fig. 48 experiment_8_act_2026-04-12_15-55-44 · train/dataloading_s

experiment_8_act_2026-04-13_22-10-30 aic-act-policy · 200 steps logged · final train/loss 0.02597

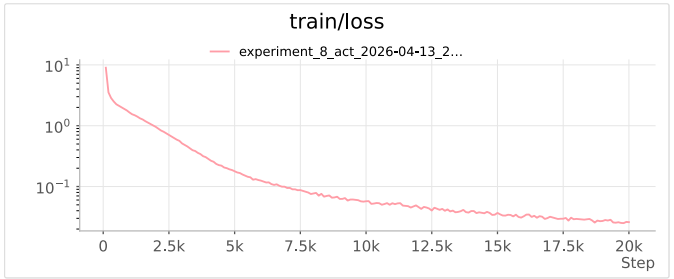


Fig. 49 experiment_8_act_2026-04-13_22-10-30 · train/loss

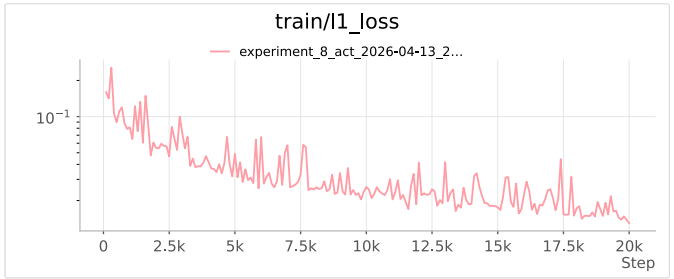


Fig. 50 experiment_8_act_2026-04-13_22-10-30 · train/l1_loss

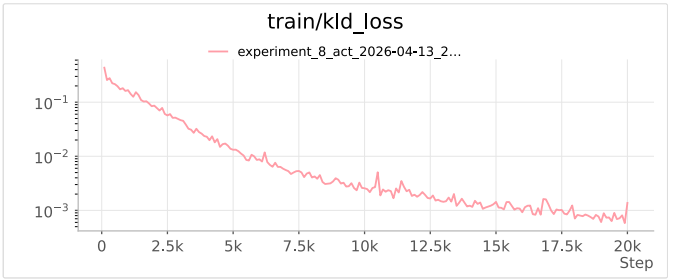


Fig. 51 experiment_8_act_2026-04-13_22-10-30 · train/kld_loss

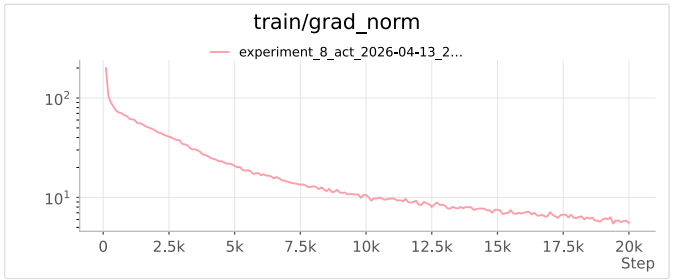


Fig. 52 experiment_8_act_2026-04-13_22-10-30 · train/grad_norm

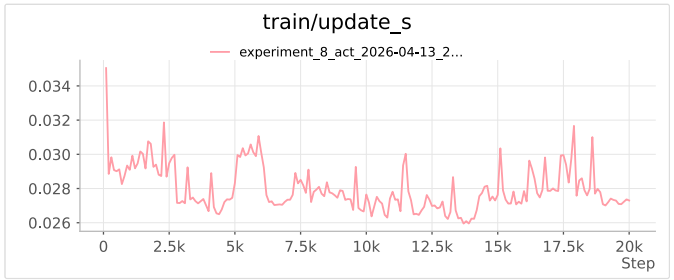


Fig. 53 experiment_8_act_2026-04-13_22-10-30 · train/update_s

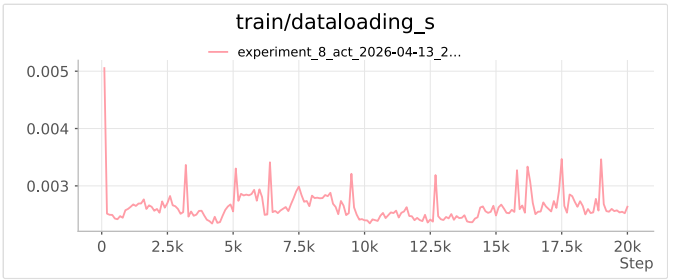


Fig. 54 experiment_8_act_2026-04-13_22-10-30 · train/dataloading_s

4 Configuration

Shared across all 8 runs

Key	Value
policy.type	act
policy.vision_backbone	resnet18
policy.pretrained_backbone_weights	ResNet18_Weights.IMAGENET1K_V1
policy.dim_model	512
policy.n_heads	8
policy.dim_feedforward	3200
policy.n_encoder_layers	4
policy.n_decoder_layers	1
policy.n_vae_encoder_layers	4
policy.use_vae	True
policy.latent_dim	32
policy.kl_weight	10
policy.chunk_size	100
policy.n_action_steps	100
policy.n_obs_steps	1
policy.dropout	0.1
optimizer.type	adamw
optimizer.lr	1e-05
optimizer.weight_decay	0.0001
optimizer.grad_clip_norm	10
scheduler	.
seed	1000
dataset.image_transforms.enable	False
dataset.use_imagenet_stats	True
use_rabc	False

Per run

Project	Run	batch_size	steps	num_workers	save_freq	dataset.repo_id	policy.pretrained_path
aic-act-training	experiment_8_act_2026-04-13_23-28-27	8	20000	8	5000	experiment_8_three_camera_images	·
	sfp_clean_198_2026-04-20_14-06-27	4	20000	2	5000	sfp_clean_198_2026-04-20_12-31-19	·
	sfp_clean_198_2026-04-20_14-20-08	4	220000	2	10000	sfp_clean_198_2026-04-20_12-31-19	·
	sfp_clean_246_2026-04-20_18-24-56	4	220000	2	10000	sfp_clean_198_plus_48_2026-04-20_16-37-48	·
	sfp_student_clean_497_48ep_2026-04-21_15-28-25	8	2640000	10	20000	sfp_student_clean_497_seed_6531465314	·
	sc_stage_fixed_scene_curated_20260508_002_terminal_v1_guarded_insert_no_imagenet_2026-05-10_13-05-24	8	20000	2	5000	guarded_insert	guarded_insert_010000_no_imagenet
aic-act-policy	experiment_8_act_2026-04-12_15-55-44	8	20000	8	5000	experiment_8_center_camera	·
	experiment_8_act_2026-04-13_22-10-30	8	20000	8	5000	experiment_8_three_camera_images	·